21. Construct the cost of living Index number for 2015 on the basis of 2012 from the following data using family budget method.

Commodity	Price		Weights
	2012	2015	
Rice	250	280	10
Wheat	70	85	5
Corn	150	170	6
Oil	25	35	4
Dhal	85	90	3

22. Calculate the cost of living index by aggregate expenditure method:

Commodity	Weights 2010	Price (Rs.)	
		2010	2015
P	80	22	25
Q	30	30	45
R	25	42	50
S	40	25	35
T	50	36	52

9.3 Statistical Quality Control (SQC)

Introduction

It is one of the most important applications of statistical techniques in industry. The term Quality means a level or standard of a product which depends on Material, Manpower, Machines, and Management (4M's). Quality Control ensures the quality specifications all along them from the arrival of raw materials through each of their processing to the final delivery of goods. This technique is used in almost all production industries such as automobile, textile, electrical equipment, biscuits, bath soaps, chemicals, petroleum products etc.

9.3.1 Meaning

Quality Control is a powerful technique used to diagnose the lack of quality in any of the raw materials, processes, machines etc… It is essential that the end products should possess the qualities that the consumer expects from the manufacturer.

9.3.2 Causes of Variation

There are two causes of variation which affects the quality of a product, namely

1. Chance Causes (or) Random causes

2. Assignable Causes

Chance Causes

These are small variations which are natural and inherent in the manufacturing process. The variation occurring due to these causes is beyond the human control and cannot be prevented or eliminated under any circumstances. The minor causes which do not affect the quality of the products to an extent are called as Chance Causes (or) Random causes. For example Rain, floods, power cuts, etc…

Assignable Causes

The second type of variation which is present in any production process is due to non-random causes. The assignable causes may occur in at any stage of the process, right from the arrival of the raw materials to the final delivery of the product. Some of the important factors of assignable causes are defective raw materials, fault in machines, unskilled manpower, worn out tools, new operation, etc.

The main purpose of SQC is to device statistical techniques which would help in elimination of assignable causes and bring the production process under control.

9.3.3 Process Control and Product Control

The main objective in any production process is to control and maintain a satisfactory

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